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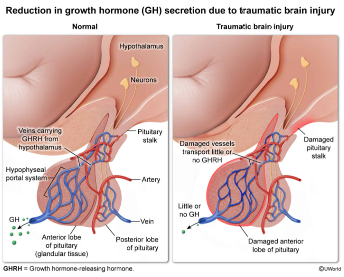
In the TBI patient described in the passage, the likely cause of the decreased GH secretion is damage:

- A. affecting cells in the pancreas that produce somatostatin. [4%]
- ✔ B. affecting cells in the pituitary stalk and anterior pituitary gland. [78%]
- C. to gut cells producing a stimulator of GH release. [2%]
- D. to glandular and neuronal cells in the posterior portion of the pituitary gland. [14%]

Correct
78% answered correctly

Explanation:



Human growth is regulated by numerous factors (eg, genes, nutritional status, pre-birth care). Multiple hormones (eg, thyroid hormone, sex hormones) influence growth, and one of the effects of GH, as its name suggests, is to stimulate the growth of most cell types.

GH secretion is highest in childhood and adolescence and declines sharply as linear growth (ie, getting taller) slows and ceases in early adulthood. Despite this decline, GH's effect on other processes (eg, metabolism) continues throughout adulthood.

The passage describes a patient who received a TBI and has decreased secretion of GH, and the question asks for the likely cause of this decrease. In response to the administration of arginine and GHRH, an endogenous stimulator of GH release by anterior pituitary cells, the patient's GH levels did not rise appreciably. This demonstrates that even when potential damage to the hypothalamus was bypassed by the infusion of exogenous GHRH, the pituitary was unable to respond by releasing GH. Therefore, **damage to** the **pituitary stalk** and **anterior pituitary** are **consistent with** the **results of** the exogenous GHRH **stimulation test**.

(Choice A) Somatostatin is produced in several locations in the body, including the hypothalamus and pancreas. The passage states that somatostatin *inhibits* GH secretion. Therefore, damage to somatostatin production in the pancreas cells would likely increase, *not* decrease, GH secretion.

(Choice C) Although GH secretion can be stimulated by hormones released from the gut, the finding that GH did not rise when GHRH and arginine were administered suggests that direct damage to GH-producing cells or to blood vessels leading to them likely occurred as a result of the TBI.

(Choice D) GH is produced by the anterior (*not* the posterior) pituitary.

Educational objective:
GH is released from the anterior pituitary gland. GHRH, carried from the hypothalamus to the pituitary gland via connecting veins, is one of the primary stimulators of GH secretion.

Time spent:0 sec
Subject:
Biology

QID:404189
System:
Endocrine and Nervous Systems

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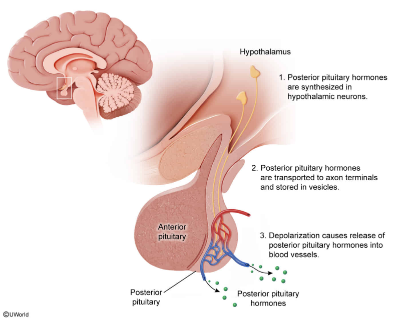
Pituitary hormones can be released into the blood from either the anterior or posterior pituitary gland. In the absence of an injury, posterior pituitary hormones are released via:

- A. endocrine cell secretion into the blood entering the posterior pituitary. [6%]
- B. neural cell secretion into the blood entering the posterior pituitary. [8%]
- C. endocrine cell secretion into the blood leaving the posterior pituitary. [60%]
- ✔ D. neural cell secretion into the blood leaving the posterior pituitary. [24%]

Correct
24% answered correctly

Explanation:

Posterior pituitary hormones are released from neurons into the blood that leaves the posterior pituitary



The pituitary gland is a small structure situated near the hypothalamus. Blood vessels and nerves pass from the hypothalamus through the pituitary stalk before entering the anterior or posterior lobes of the pituitary gland.

The two pituitary lobes are both derived from embryonic ectodermal tissue but differ in composition. The anterior pituitary is derived from epithelial cells from the developing roof of the mouth and contains typical glandular endocrine cells, whereas the posterior pituitary is derived from ectodermal neural tissue in the developing brain and does not contain typical glandular endocrine cells.

The passage describes how hormone release can be altered in individuals who have suffered a TBI, and the question asks about the process of posterior pituitary hormone secretion in the absence of injury. As noted above, the posterior pituitary does not contain typical glandular endocrine cells but is derived from neural tissue.

The **posterior pituitary** is **a storage site for neurohormones** (ie, hormones released from neurons rather than from typical glandular endocrine cells) produced by hypothalamic neurons whose axons extend down into the posterior pituitary. When stimulated, these neurohormones are **released** by exocytosis from the **axons into** the small **blood vessels** that **carry blood away** from the **posterior pituitary**.

(Choices A, B, and C) Posterior pituitary hormone release is from *neural cell* axons into the small blood vessels leaving (*not* entering) the posterior pituitary.

Educational objective:
Posterior pituitary hormones are made in the hypothalamus and stored within axons projecting into the posterior pituitary. When stimulated, these axons release the stored hormones into the blood vessels leaving the posterior pituitary.

Time spent:0 sec
Subject:
Biology

QID:404190
System:
Endocrine and Nervous Systems

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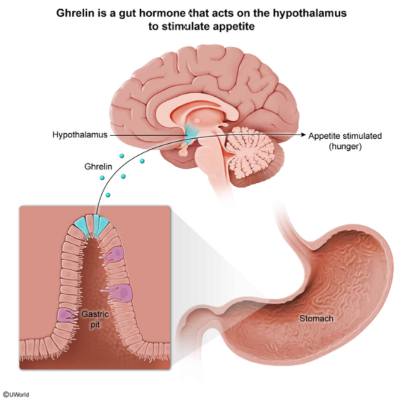
During the GH secretion test, GH-releasing hormone (GHRH) was administered to the patient along with arginine. GHRH is an endogenous stimulator of GH secretion, and arginine inhibits secretion of somatostatin, an endogenous inhibitor of GH secretion. When administered together in healthy individuals, GHRH and arginine strongly stimulate GH secretion into the blood. However, this patient exhibited an abnormally small response to GHRH and arginine, leading the physician to start the patient on GH therapy. After a year, the patient's symptoms had improved, and body composition testing showed that her lean body mass had increased.

Which gut-derived hormone would promote appetite in a healthy individual who has not had a TBI?

- A. Cholecystokinin [3%]
- B. Leptin [18%]
- ✔ C. Ghrelin [76%]
- D. Epinephrine [1%]

Correct
75% answered correctly

Explanation:



Sufficient energy and essential nutrients must be taken in via the diet to supply the many energy- and nutrient-demanding processes occurring in the body (after accounting for the inherent inefficiency of these processes). If dietary intake is chronically too low to meet the body's metabolic demands, body mass will be lost as endogenous molecules are broken down to compensate for deficient intake of energy and essential molecules. With sustained deficiency, function may be compromised. However, chronic excessive intake can also compromise function by increasing metabolic costs and susceptibility to certain metabolic or body composition–related disorders (eg, insulin resistance, type 2 diabetes mellitus).

The balance between intake and utilization of dietary nutrients is regulated by numerous factors, including hormones released from cells throughout the body (eg, intestine, adipose tissue, pancreas). **Some** of these **hormones promote** desire for **food intake** (ie, appetite), and **others promote** a feeling of **satiety** (ie, fullness or dietary satisfaction).

The question asks for the gut-derived hormone that would promote appetite in a healthy individual who has not had a TBI. **Ghrelin** is a hormone **released by** cells in the **stomach** (ie, part of the gut). In a healthy individual, ghrelin is transported via the bloodstream to the brain, where it **acts on** the **hypothalamus to stimulate appetite**.

(Choice A) Cholecystokinin is a gut-derived hormone, but it promotes satiety rather than appetite.

(Choices B and D) Leptin and epinephrine are not gut-derived hormones.

Educational objective:
Ghrelin is a hormone released from stomach cells that stimulates appetite by acting on cells in the hypothalamus.

Time spent: 0 sec
Subject: Biology

QID: 404191
System: Endocrine and Nervous Systems

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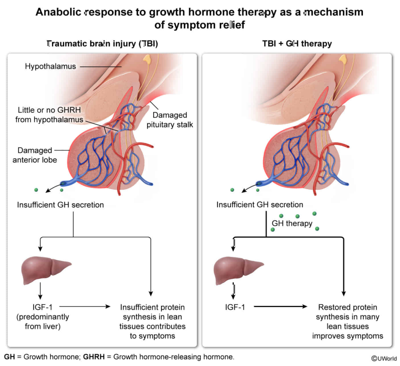
Based on the passage, a likely mechanism by which GH therapy relieves chronic symptoms associated with TBI is:

- A. stimulating protein synthesis in multiple tissues. [56%]
- B. inhibiting protein synthesis in multiple tissues. [2%]
- C. stimulating GHRH release by the pituitary. [38%]
- D. inhibiting GHRH release by the pituitary. [3%]

Correct
56% answered correctly

Explanation:



Proteins play numerous essential roles in organisms. Growth, development, and repair of tissues require the synthesis of new proteins, and numerous stimuli (eg, mechanical stress, nutrition, inflammation) influence protein synthesis. Stimuli that promote the accumulation of newly synthesized proteins are referred to as anabolic, whereas stimuli that inhibit protein synthesis (or stimulate protein degradation) are referred to as catabolic.

Hormones exert a significant influence over tissue growth and remodeling, partly through their influence on protein synthesis. In the context of tissue growth and remodeling, anabolic hormones are those associated with the accumulation of newly synthesized proteins, whereas catabolic hormones are those associated with decreased protein accumulation.

The passage describes a scenario in which an individual suffered a TBI and underwent testing, the results of which suggested that the individual's capacity for GH production was abnormally low. The question asks for a likely mechanism by which GH therapy might relieve symptoms associated with TBI. GH is an anabolic hormone, which means that it exerts a positive effect on protein synthesis and cell growth in numerous tissues. Therefore, one mechanism by which GH therapy might provide symptom relief is through **stimulation of protein synthesis** in multiple tissues.

(Choice B) Anabolic hormones like GH typically stimulate (*not* inhibit) protein synthesis.

(Choices C and D) GHRH is released from the hypothalamus (*not* the pituitary).

Educational objective:
GH is an anabolic hormone that promotes protein synthesis and growth in multiple tissues.

Time spent: 0 sec
Subject: Biology

QID:404192
System: Endocrine and Nervous Systems

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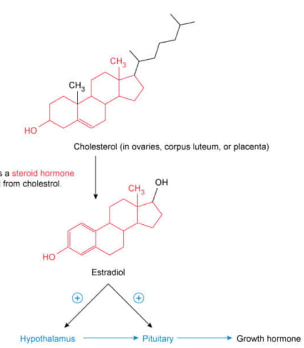
Based on the passage, does estradiol act as a steroid hormone, a tropic hormone, or both? Estradiol acts as:

- A. a steroid hormone. [39%]
- B. a tropic hormone. [9%]
- ✔ C. both a steroid hormone and a tropic hormone. [49%]
- D. neither a steroid hormone nor a tropic hormone. [1%]

Correct
49% answered correctly

Explanation:

Estradiol is both a steroid hormone and a tropic hormone



Estradiol is a tropic hormone stimulating growth hormone release via effects on the hypothalamus and pituitary.

Hormones are signaling molecules that are secreted in relatively small amounts from endocrine tissues into the circulation. Despite their low concentration, hormones can cause profound responses when they bind to their receptors in target cells. Hormones can be classified according to their characteristics using several schemes.

One hormonal classification scheme is based on structure and divides hormones into three broad categories:

- **Peptide hormones** are small proteins and, as such, consist of amino acids linked via peptide bonds.
- **Steroid hormones** are lipids derived from cholesterol that has been modified via a series of chemical reactions.
- **Amino acid–derived hormones** are molecules derived from either tyrosine or tryptophan; these hormones are sometimes grouped together with peptide hormones, as all are derived from amino acids.

Another scheme for classifying hormones is based on their influence on other hormones. In this system, hormones that can target endocrine tissues to influence the secretion of another hormone, and thereby contribute to a specific response, are designated as tropic hormones. Hormones that act directly on target cells to elicit nonendocrine responses are called direct hormones in this scheme. Some hormones act as both direct and as tropic hormones.

The passage describes a scenario in which TBI is associated with abnormally low GH secretion and notes that estradiol influences GH secretion. The question asks whether estradiol acts as a steroid hormone and/or a tropic hormone. Because its carbon skeleton is **derived from cholesterol**, **estradiol is** a lipid-soluble **steroid hormone**, and because it **influences GH** secretion, estradiol is also a **tropic hormone**.

(Choices A, B, and D) Estradiol is both a steroid sex hormone and a tropic hormone.

Educational objective:
Estradiol is a steroid hormone derived from cholesterol, and in addition to acting directly on nonendocrine target tissues, it can function as a tropic hormone.

Time spent:0 sec
Subject:
Biology

QID:404193
System:
Endocrine and Nervous Systems

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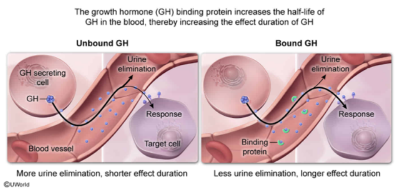
Based on the passage, GH is bound to a binding protein, which most likely influences responses to activation of the GH axis by:

- A. decreasing the availability of GH for binding to its intracellular receptor. [13%]
- B. increasing the availability of GH for binding to its intracellular receptor. [36%]
- C. decreasing the effect duration of GH. [1%]
- ✔ D. increasing the effect duration of GH. [48%]

Correct
47% answered correctly

Explanation:



After secretion into the blood, hormones may have several fates, including modification, binding to transport proteins, binding to receptors in target tissues, destruction, and elimination. The response to a hormone is dependent upon the balance between these various fates. For example, some hormones (eg, insulin) are internalized and degraded after receptor binding in target tissues, thereby terminating the effect of individual hormone molecules.

The passage states that some of the GH secreted into the circulation becomes bound to a binding protein and that this binding increases GH's half-life in the circulation. The question asks for the most likely influence of this binding on responses to activation of the GH axis. An **increase in** circulating **half-life corresponds to** a **longer persistence**; therefore, **bound GH** will be **present** for a **longer** time **in** the **blood**, which **will likely increase** the **effect duration**.

(Choices A and B) GH is a peptide hormone. In general, peptide hormones activate receptors located on cell membranes because peptide hormones are unable to cross the phospholipid bilayer of target cells (in contrast, steroid hormones typically have intracellular receptors).

(Choice C) An increased half-life in the blood due to reduced filtration by the kidney would increase (not decrease) the effect duration of GH.

Educational objective:
The response elicited from a target tissue by the binding of a circulating hormone to its receptor is modulated by factors influencing the hormone's availability at the target tissue (eg, secretion, transport protein binding, degradation/elimination).

Time spent:0 sec
Subject:
Biology

QID:404194
System:
Endocrine and Nervous Systems